

International Journal of Crashworthiness

ISSN: (Print) (Online) Journal homepage: https://www.tandfonline.com/loi/tcrs20

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To cite this article: Khanh Giang Le , Pei Liu & Liang-Tay Lin (2020): Traffic accident hotspot identification by integrating kernel density estimation and spatial autocorrelation analysis: a case study, International Journal of Crashworthiness, DOI: 10.1080/13588265.2020.1826800

To link to this article: https://doi.org/10.1080/13588265.2020.1826800

Published online: 29 Oct 2020.

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INTERNATIONAL JOURNAL OF CRASHWORTHINESS
https://doi.org/10.1080/13588265.2020.1826800

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Traffic accident hotspot identification by integrating kernel density estimation and spatial autocorrelation analysis: a case study

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ABSTRACT

The study presented an approach that integrated kernel density estimation (KDE) algorithm and spatial autocorrelation analysis, which helped to determine traffic accident (TA) hotspot locations and simultaneously evaluate the statistical significance of the hotspot clusters. Firstly, hotspots were identified by applying a GIS-based KDE algorithm. Secondly, the hotspot clusters were evaluated in terms of statistical significance by applying the Moran's I statistic indices. Finally, hotspots were arranged according to their significance. TA data in the (2015–2017) period in Hanoi, Vietnam were applied to test this approach. Importantly, the study proposed a validation process for the results by applying the Gi* statistics method to validate the (H-H) clusters and the EPDO method to validate the ranking of hotspots. The results showed that this integration overcame the drawbacks of the KDE method. The statistical test process of clusters helped to hinder the occurrence of too many clusters that were determined by the KDE method because they were not really dangerous. This approach was helpful and precise in identifying TA hotspots with the statistical meaning. These outcomes will not only enable traffic authorities to comprehensively understand the reasons for each accident but also to help them manage and deal with hazardous areas according to the prior order in case of limited expenses and allocate traffic safety sources accordingly.

ARTICLE HISTORY
Received 13 April 2020
Accepted 15 September 2020

KEYWORDS
Traffic accidents (TA); hotspot; geographic information system (GIS); kernel density estimation (KDE); spatial autocorrelation; statistical significance

1. Introduction

Traffic accidents (TA) are currently a matter of primary concerns in many countries. According to the World Health Organization (WHO), there were about 1.35 million deaths and 50 million injured owing to TA every year around the world. Trauma due to TA is the leading reason for death for people aged from 5 to 29 years old [1]. In order to significantly reduce TA, it is crucial to identify locations where accidents frequently occur. These locations are often called TA black spots or hotspots [2,3]. This is the first important step in traffic safety strategy includes four phases: determination, classification, profile, and treatment [4,5]. From there, more detailed investigations will be conducted to answer questions about why and what makes these accidents happened. From there a safe driving environment will be built.

There are many methods that enable us to determine TA hotspot locations. Past studies showed that these methods can be grouped into two main groups such as non-spatial and spatial models [6]. The non-spatial models have several drawbacks, including: First, they cannot visually identify hotspots, which make it difficult to manage and overcome the hotspots; Second, Miaou and Song [7] suggested that the spatial correlation issue should be mentioned in identifying hotspots because it dramatically affects the goodness of fit of model. The spatial models can deal with both these issues [8]. Geographic information system (GIS) is a very powerful tool for analyzing TA and has been broadly used in detecting TA hotspots in many nations. GIS can visualize the locations of accidents and store its attributes. It means the dangerous locations are displayed correctly on a map linked with the accident's attribute information. Therefore, the reasons behind each accident are easy to investigate [3,8–10]. This enables traffic authorities to easily manage hotspots and promptly propose effective countermeasures.

Many previous studies showed that kernel density estimation (KDE) algorithm has been widely used in identifying TA dangerous locations effectively [8,11–14]. In comparison with other clustering techniques including k-means, k-medoids, CURE, BIRCH, STING, CLIQUE, DBSCAN, and OPTICS, the KDE algorithm has many outstanding advantages [15]. Firstly, this algorithm helps to identify the spread of the hazard level of an accident. The expansion of the hazard level can be described as a zone nearby an identified hotspot where there is a raising possibility for an accident to happen according to the spatial relationship. Secondly, an arbitrary spatial analysis unit will be characterized and identical for the entire region that creates a comparison and then a classification [4]. Thirdly, a raster layer with pixels that are assigned values corresponding to event intensity is created as a result of a KDE analysis process. Thus, the KDE method is effective for visualization because regions of high accident density can be distinctly determined [16].

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However, the KDE method lacks an evaluation of the statistical significance of the determined hotspots. This is the major disadvantage of the KDE [4,14,17]. The KDE method is more suitable for visualizing than for determining hotspot locations [18]. It is necessary to develop the statistically significant evaluation methods for the outputs of the KDE method [4,14]. However, in many previous studies, statistically significant investigation issue of hotspots determined by the KDE method was often ignored.

From the literature review process, there are very few studies that mentioned this issue. First [8], used several traditional statistical techniques such as Poisson distribution and Chi-square test. However, the results were not objective. Second, in the study of Bíl et al. [19], this issue was mentioned. This study used the network KDE method to identify hotspots and Monte Carlo method to test statistical significance. Besides, this study computed the strength and stability of the clusters. However, this study only focused on road sections excluding urban road areas and at intersections as well at highways. The drawback of this method is that it cannot identify the dangerous segments in which there are many accidents happened in case of evenly distributed. Another study of Xie and Yan [14] that integrated the network KDE and local Moran's I to determine hotspots in an urban zone and test statistical significance of the hotspots. Nevertheless, some researchers [20,21] recommended that the planar KDE should be applied for urban area because it suitable for dense networks with dispersed TA. In addition, the study of Xie and Yan [14] only used a 100-m search radius. Thus, it is necessary to have more sensitivity analyses applying other parameter values. Also of importance, accident severity index (SI) was not considered in these studies. According to many researchers, it is better to take into account injury severity in an accident outcome because the effects of accidents may be changing in relation to their associated injuries [12,22].

The aim of our study was to evaluate the statistical significance of TA hotspots that were determined by KDE method in case of taking into account SI. Our study focused on an urban area. Hanoi is the capital of Vietnam, where very complicated accidents have occurred, as a case study for our approach. Thus, firstly, the planar KDE method was applied to determine the TA hotspot locations. Secondly, the statistical significance of the hotspots was evaluated by applying global and local Moran's I methods. Finally, their orders were organized according to their dangerous levels. The results were then validated to equivalent property damage only (EPDO) method and Getis-Ord Gi* statistics. The remainders of the paper are organized as follows. Section 2 depicts proposed methodology. Section 3 illustrates the analysis of a case study. The validation of the results was implemented in Section 4. Traffic accident preventive strategies are showed in Section 5. Conclusions are showed in Section 6.

2. Methodology

In this study, the planar KDE algorithm was applied to identify TA hotspots. However, this method lacks an evaluation of the statistical significance of the hotspots as mentioned above. Therefore, this study proposed a procedure aiming to improve the effectiveness and accuracy of the KDE method. Figure 1 presents the combination of the KDE method with a statistical significance evaluation process of the resulting clusters. The proposed methodology was carried out by the following steps:

Initially, TA locations were geocoded on the digital road network. However, TA locations may be reported at the same locations. Therefore, collect events tools were applied to integrate and collect accidents that occurred at the same locations. This step created a collected events map. In addition, accident SI was considered in determining TA hotspots because the accident SI determines the dangerous level of an accident. Subsequently, the planar KDE algorithm was applied to calculate and create a TA density map.

On the other hand, in order to carry out the statistically significant evaluation process, the main steps were implemented as follows. First, it is necessary to test the random distribution of TA in a section. If the accidents are distributed randomly, this process stops. Global statistics was applied in this step. Conversely, if the accidents are non-random distribution, it is necessary to determine bandwidth in which autocorrelation or clustering phenomena is maximized. To do this, incremental global Moran's I was applied and this process was repeated many times. However, what a starting distance at which any given point has at least one neighbour needs to be found out. After getting the optimal bandwidth, this bandwidth will be the threshold distance input. Then, a significant map of clusters was generated by applying local Moran's I. Then, a TA hotspots priority map was produced as a result of the combination between the TA density map and the significant map of clusters. Finally, to validate the proposed methodology, the obtained results were compared to the Getis-Ord Gi* statistics and the EPDO method.

The following sections will explain in more detail about the methods applied in this study including the planar KDE, the global and local Moran's I statistics as well as the Gi* and the EPDO method.

2.1. Severity index (SI)

It is vital to take into account injury severity in an accident outcome because the effects of accidents may change in relation to their associated injuries. The accident SI measures the seriousness of an accident. Higher weights are assigned to higher levels of injuries rely on the rate of their average accident expenditure to property damage only (PDO) accidents [22]. This study used a crash severity weight system computed based on Vietnam's conditions. SI for each location can be calculated as follows:

$$SI = 10\ I + 40\ D \quad (1)$$

where SI is severity index for each location; I is the total number of injuries; and D is the total number of deaths.

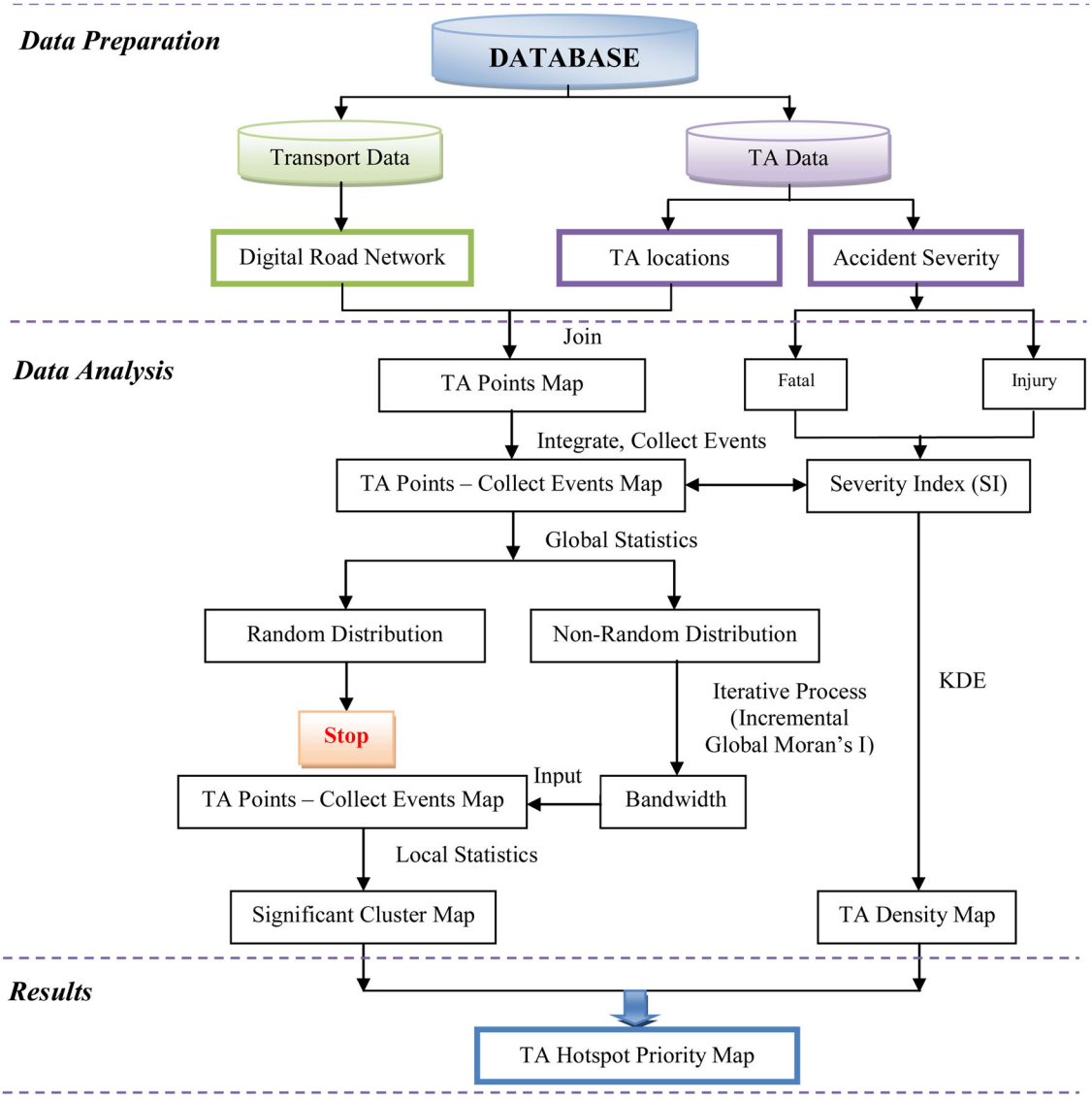


Figure 1. The flowchart of statistically significant evaluation process of the hotspots.

2.2. Planar kernel density estimation

There are several spatial analysis tools that enable us to comprehensively understand the geographical change of point models. The planar KDE method has long been widely applied to identify accident hotspots [10,12,13]. A circular search zone is determined by a kernel function and put over each accident. This generates a smooth surface. Later, a network of cells is overlaid over the research zone. Then, a kernel function is used that goes from 1 at the centre point of the event to 0 at the radius of the search area [4]. The intensity at a location calculated by Equation (2) [23].

$$f(s) = \sum_{i=1}^{n} \frac{1}{\pi r^2} k\left(\frac{d_{is}}{r}\right) \tag{2}$$

where $f(s)$ is the intensity at the position s, r is the search radius, k is the kernel function, and d_{is} is the distance among s and i^{th}.

The output of the planar KDE method is presented in a raster format consisting of a grid of cells. Many researchers stated that the selection of the bandwidth r is more important than the selection of the kernel function k [16]. There are three kernel functions popularly applied including Gaussian function, Quartic function, and Minimum variance function [24]. The Gaussian function was selected in our study and presented as:

$$k\left(\frac{d_{is}}{r}\right) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{d_{is}^2}{2r^2}\right), \text{ when } 0 < d_{is} \le r \tag{3}$$

$$k\left(\frac{d_{is}}{r}\right) = 0, \text{ when } d_{is} > r \tag{4}$$

2.3. Global Moran's I statistic

The statistically significant evaluation process of the hotspots was started by testing the spatial autocorrelation of point events. It is necessary to test the random distribution of accidents. If the accidents are the random distribution, this process stops. Conversely, the process is continued.

The global Moran's I tests spatial autocorrelation based on positions and attributes of the features simultaneously

[25]. Our study applied the global Moran's I as the first spatial autocorrelation evaluation method. The global Moran's I index is expressed by Equation (5):

$$I = \frac{n}{S_0} \frac{\sum_{i=1}^{n} \sum_{j=1}^{n} w_{i,j} z_i z_j}{\sum_{i=1}^{n} z_i^2} \quad (5)$$

where z_i, z_j are the deviations from the mean; w_{ij} is the spatial weight between feature i and j; n is the quantity of features, S_0 is the aggregate of all the spatial weights and calculated as follows:

$$S_0 = \sum_{i=1}^{n} \sum_{j=1}^{n} w_{i,j} \quad (6)$$

The z_I -score is calculated as follows:

$$z_I = \frac{I - E[I]}{\sqrt{V[I]}} \quad (7)$$

where

$$E[I] = -1/(n-1) \quad (8)$$

$$V[I] = E[I^2] - E[I]^2 \quad (9)$$

The global Moran's I values vary from −1 to 1. A value of 1 indicates great positive spatial autocorrelation, a value of −1 indicates great negative spatial autocorrelation, and a value of 0 shows great spatial randomness [26].

2.4. Local Moran's I statistic

While global Moran's I investigates the spatial autocorrelation as a whole, a local indicator of spatial association (LISA) investigates the levels of spatial autocorrelation at each particular position [27]. This was the second method that determined local clusters and outliers. Furthermore, LISA can be visualized as a map. The local Moran's I index is expressed as Equation (10):

$$I_i = z_i \sum_{j=1}^{n} w_{i,j} z_j \quad (10)$$

where similar to the global Moran's I, z_i and z_j are the deviations from the mean of the sample, w_{ij} is the spatial weight between feature i and j.

2.5. The validation process

According to many researchers, the Gi* statistics was commonly applied to determine whether features with high values or low values tend towards cluster [28,29]. And, the EPDO method is a unique method in network screening methods that takes into account SI in determining hotspots [30,31]. Therefore, the Gi* statistics was applied to validate the (H-H) clusters and the EPDO method was applied to validate the ranking of hotspots.

2.5.1. Gi* statistics method

The locations of (H-H) clusters were validated by the Gi* method. The outputs of this method include Gizscores and Gipvalues that show whether features with high values or features with low values tend towards cluster. In case a feature with a high value is surrounded by others with high values (hotspots), the hotspot is statistical meaning. Gi* statistic is a z-score, calculated as follows [32] :

$$G_i^* = \frac{\sum_{j=1}^{n} w_{i,j} x_j - \overline{X} \sum_{j=1}^{n} w_{i,j}}{S \sqrt{\frac{\left[n \sum_{j=1}^{n} w_{i,j}^2 - \left(\sum_{j=1}^{n} w_{i,j} \right)^2 \right]}{n-1}}} \quad (11)$$

where x_j is feature j's the attribute value; n is the number of features, and:

$$\overline{X} = \frac{\sum_{j=1}^{n} x_j}{n} \quad (12)$$

$$S = \sqrt{\frac{\sum_{j=1}^{n} x_j^2}{n} - (\overline{X})^2} \quad (13)$$

A positive z-score (>1.96) indicates the observations are spatial clusters (hot spots) and conversely (cold spots). The p-values are suggested to examine the significance of Gi*. In case of z-score >1.96 and p-value < 0.05, the observations are spatial clusters and Gi* is statistical meaning at the confidence level of 95%. Conversely, in case of p-value ≥ 0.05, the observations are random.

2.5.2. Equivalent property damage only (EPDO) method

The ranking of hotspots was validated by the EPDO method. Social accident costs are applied to compute the EPDO weights based on the willingness to pay approach [30,33]. In Vietnam's situations, the average accident costs were applied as Table 1.

The steps are carried out as follows [30]:

Step 1: The weighting factors are computed as follows:

$$f_{y(weight)} = \frac{CC_y}{CC_{PDO}} \quad (14)$$

where: $f_{y(weight)}$ is weighting rely on accident severity, y; CC_y is accident expenditure for accident severity, y; CC_{PDO} is accident expenditure for PDO accident severity.

The outputs of calculating the weighting factors were also presented in Table 1.

Step 2: Compute EPDO scores:

Table 1. The average accident costs and the weights for various accident severities in Vietnam.

	Average Accident Cost (2006)		
Severity	VND	USD	Weight
Fatal (F)	2000000000	86000	40
Injury (I)	500000000	21000	10
PDO (P)	50000000	2100	1

(Source: Accident cost estimates by damaged levels (Decision No.768/2006/QD-BCA)).

The frequency of PDO, injury, and fatal accidents is depended on the number of accidents, not the number of injuries per accident. For each intersection or segment, EPDO scores are computed as follows:

$$Total\ EPDO\ Score = f_F N_{ob,i(F)} + f_I N_{ob,i(I)} + f_P N_{ob,i(P)} \quad (15)$$

where: f_F, f_I, f_P are accident weights corresponding to fatal, injury, and PDO accidents; $N_{ob,i(F)}$, $N_{ob,i(I)}$, $N_{ob,i(P)}$ are number of accidents per intersection or segment, i, corresponding to fatal, injury, and PDO accidents.

Step 3: Rank locations:

The EPDO scores were used to rank the locations in descending order.

3. Analysis of a case study

3.1. Study area and data

3.1.1. Study area

This research was performed in Hanoi that is located in the northern area and the capital of Vietnam (Figure 2(a)). Hanoi covers an area of 3,328.9 km^2 with 8.0 million people as of 2019, and about 20,000 km of road system. The primary transportation means in Hanoi are motorbikes (86%), buses, taxis, and an increasing quantity of private cars in recent years. Hanoi has the highest growth rate of personal vehicles in Vietnam with more than 6.6 million vehicles that are in circulation and more than 30,000 new registrations each month.

The process of urbanization has been taking place in Hanoi with a fast speed leading to an increase in population accompanied by an increase in the number of means of transportation in recent years as the main cause of rising traffic congestion and accidents. The reason is that roads and infrastructure system in old areas of Hanoi are no longer suitable for the current travel situation. In order to respond promptly to the new situations, Hanoi has built many new roads. However, besides those formal roads, there were many roads opened illegally and not controlled by the city government. These illegal roads were opened by the citizens living in that area to facilitate their travel and business. This fact is very worrying because it is also one of the main causes of serious accidents in Hanoi as well as in Vietnam. These illegal roads formed dangerous crossings with a lack of safety equipment. These conflicting locations were also often the location of accident hotspots in Hanoi according to statistics.

There were 1,365 accidents, causing 544 deaths and 916 injured in 2018 in Hanoi. This is an alarming number. Therefore, the first countermeasure needs to be implemented to mitigate accidents in Hanoi is to identify potential accident locations (hotspots), including the illegal conflict locations mentioned above. However, the current determination of accident hotspots in Vietnam is still very backward. Vietnam uses accident frequency method to identify accident hotspots [34]. According to many previous studies, this is the oldest approach to determine hotspots. This approach has several disadvantages like lacking of visualization, spatial-temporal relationship, without SI, and lacking of priority ranking [6]. Thus, this study applied the proposed methodology to identify the TA hotspots in a case study of Hanoi.

3.1.2. Study data

There are two different data sets used for this study, including:

1. A road network map of Hanoi in a shape file format that was provided by the department of Natural Resources and Environment in Hanoi, which includes length, width, and types of road, speed limits, etc.
2. An accident data set from 2015 to 2017 in Hanoi, which was given by the department of Transport Police. This data set covers important information including the accident's date, time, and position, types of accident and vehicle, information of drivers and the injured, etc. These two data sets were combined and manipulated together by using ArcGIS software 10.2. Figure 2b illustrates the distribution of accident locations in three years (2015-2017) in Hanoi (This data set was used in a previous study [12], but with the different contents).

3.2. Results and discussion

3.2.1. Kernel density estimation

Firstly, according to the proposed methodology, after geocoding the accident locations and combining with the road network map, a map of accident locations was generated as shown in Figure 2b. Next, the KDE method was applied to create a TA density map in case of taking into account accident SI. According to the suggestion from the previous study [12], accident SI should be considered in identifying the TA hotspots. According to Anderson [4], cell size and search radius (bandwidth) are the two key parameters affecting to the KDE method. Many researchers stated that the bandwidth is the most critical criterion for identifying the most suitable density surface [35]. Thus, the result of the hotspots will be significantly affected by the selection of bandwidth. Namely, the smaller the bandwidth, the smaller the hotspots will be. The smoothness of the density surface will be affected by the value of bandwidth – the smoother the density surface, the larger the bandwidth will be. Therefore, it is vital to select an optimal bandwidth. According to many researchers, the bandwidth is often in the range (20 m to 1000 m) [23]. In this study, we applied a 1000 m-bandwidth value, because this value has been proven to be consistent with our data set in our previous studies [12,36].

The KDE method enables us visualize the TA hotspots easily. Figure 3 showed TA hotspot locations in Hanoi (2015–2017), which mainly located on Van Dien station – Ngoc Hoi road, Cho Tia, Van Diem – Thuong Tin zone, Quang Trung – Nguyen Trai intersection, and the southern portion of several bridges such as Thang Long, Nhat Tan, Long Bien, etc. However, the KDE method lacks an evaluation of the statistical significance of the determined hotspots as mentioned above. Thus, it is essential to investigate

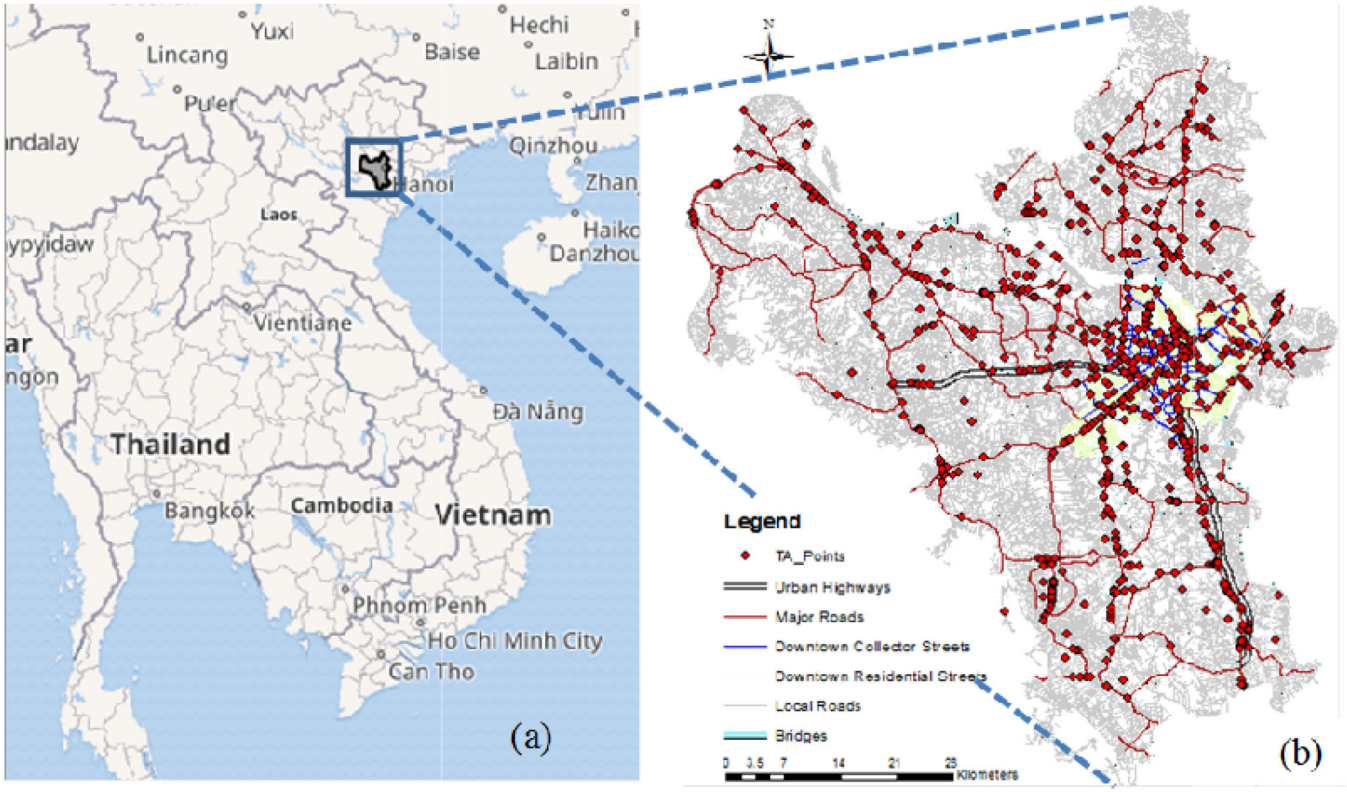


Figure 2. a) Map of Vietnam with the highlight of geographic location of Hanoi area (Source: Wikipedia, 2020); b) Study region in Hanoi with the distribution of accidents in three years (2015-2017).

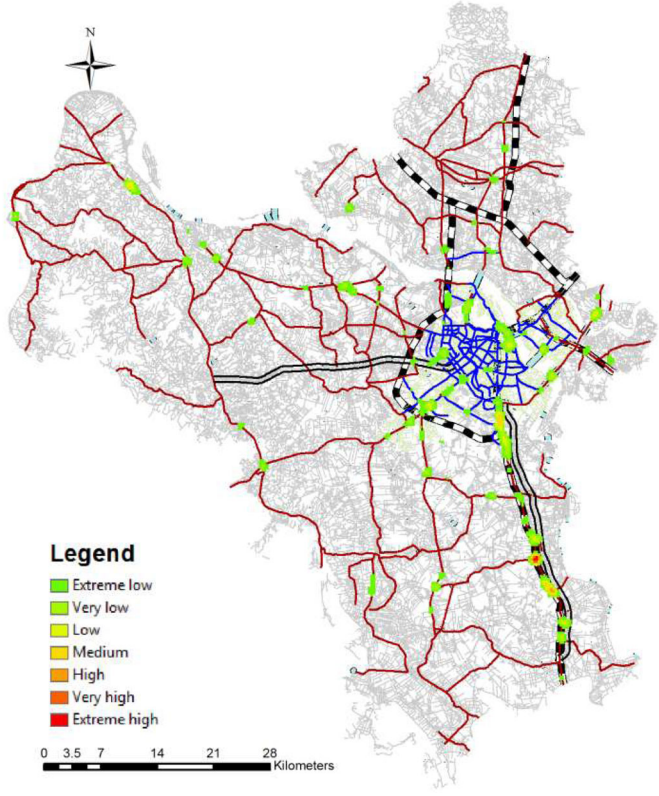


Figure 3. Map representing TA hotspot locations in Hanoi (2015–2017).

the hotspot's statistical significance, find out the most hazardous locations with statistical significance, and rank them according to their dangerous levels.

3.2.2. Statistically significant evaluation process

3.2.2.1. Global spatial autocorrelation analysis. In general, the KDE application has completed in the above step (Section 3.2.1). The KDE method determined the accident hotspot locations as shown in Figure 3. According to Erdogan et al. [8], occasionally, an arbitrary degree of statistical significance is identified by the KDE method. Nevertheless, it is better if the identified hotspot's statistical significance is determined objectively and proactively. Our process of statistically significant evaluation was by starting with the null hypothesis: "the accidents in a section occurred randomly". The null hypothesis's statistical testing was based on a spatial autocorrelation tool in ArcGIS software 10.2 that is global Moran's I.

From Figure 4, the result showed that this dataset demonstrates a clustered pattern because the value of Moran's I is 0.497, z-score of 9.99, and p-value of 0.00. Thus, we can reject the null hypothesis. Therefore, the next step is to identify a bandwidth value in which clustering phenomena is maximized. Incremental global Moran's I was applied and this process was repeated many times. However, what a starting distance at which any given point has at least one neighbour needs to be found out. Calculate distance band from neighbour count was used to compute, then we have a result is 500 m. Then, incremental spatial autocorrelation was applied to calculate and the result is shown in Figure 5 and Table 2.

The spatial autocorrelation was repeated at different distances and each distance got a z-score value that is the statistically significant level (Figure 5). It deviated from our

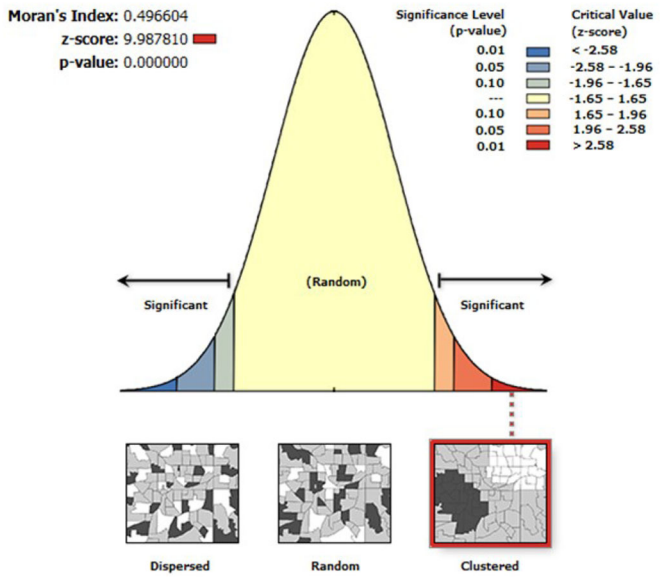


Figure 4. Spatial autocorrelation.

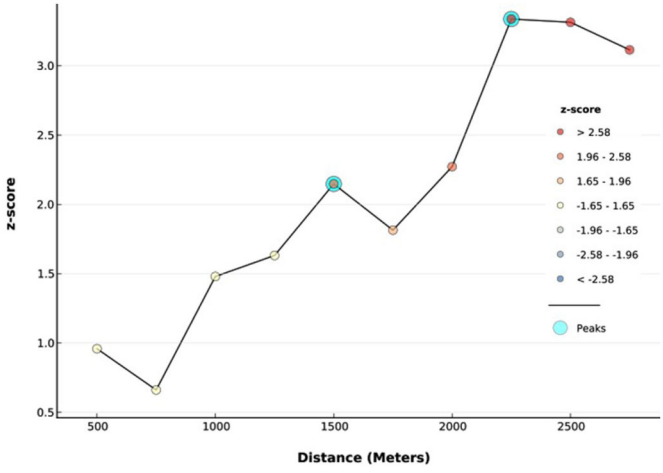


Figure 5. Spatial autocorrelation by distance.

Table 2. Global Moran's I by distance.

Distance (m)	Moran's Index	Expected Index	Variance	z-score	p-value
500.00*	0.026614	−0.001616	0.000867	0.958860	0.337629
750.00*	0.018387	−0.001372	0.000892	0.661623	0.508213
1000.00*	0.039438	−0.001307	0.000758	1.480188	0.138823
1250.00*	0.041200	−0.001233	0.000676	1.631558	0.102773
1500.00*	0.049820	−0.001202	0.000564	2.147713	0.031737
1750.00*	0.038418	−0.001182	0.000476	1.814229	0.069643
2000.00*	0.044918	−0.001168	0.000411	2.272086	0.023081
2250.00*	0.060999	−0.001156	0.000347	3.337184	0.000846
2500.00*	0.054846	−0.001152	0.000286	3.313633	0.000921
2750.00*	0.048476	−0.001147	0.000254	3.114572	0.001842

*At least one distance increment resulted in features with no neighbors which may invalidate the significance of the corresponding results.
First Peak (Distance, Value): 1500.00, 2.147713.
Max Peak (Distance, Value): 2250.00, 3.337184.

random hypothesis and when these z-score values were compared among the various increments of distances, it revealed that some of them are higher than others. Thus, there is the fact that a couple of peaks where the z-score values get very high which is an indication that it is at those distances where the clustering is maximized. We are more likely to find natural clustering within the data. In addition, Table 2 shows that the maximum peak is at 2250 m at which we can find maximum clustering and that is the number that we need in order to process forward.

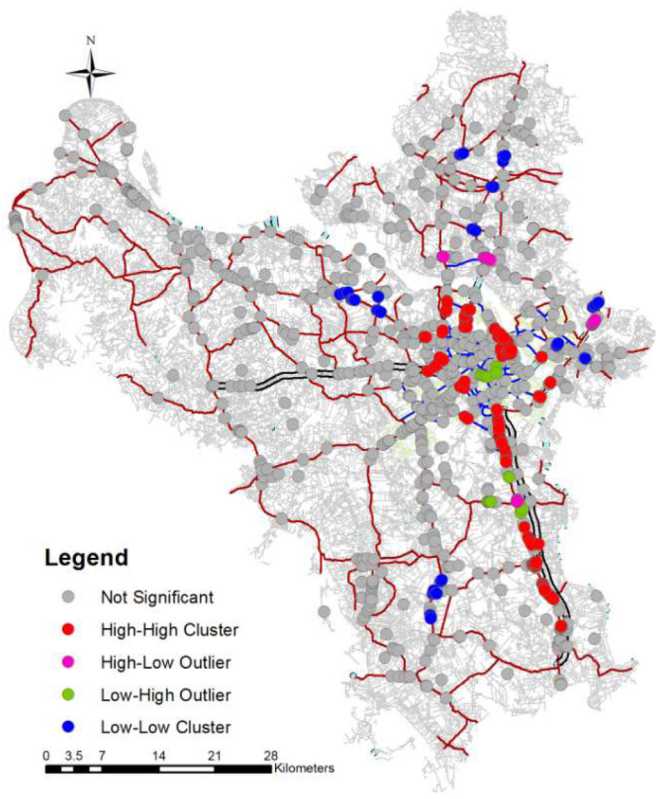


Figure 6. Map shows the significance of clusters with statistical meaning.

3.2.2.2. Local spatial autocorrelation analysis:. In the next step, the local Moran's I statistics was applied to calculate and generate a significant map of clusters with statistical meaning (Figure 6). Figure 6 showed that all the grey points were not significant because their z-score values were low and p-values were greater than 0.05, so there was nothing really happening. Generally, the correlation types include high-high (H-H) and low-low (L-L) that show a positive autocorrelation, while high-low (H-L) and low-high (L-H) indicate a negative autocorrelation [16].

The (H-H) points (130 points in our case study) are dangerous areas where a high number of accidents are surrounded by high values. So, in this case, we have gotten TA point's high priority near each other and those clusters were statistically significant meaning that they were very far from the random arrangement. In contrast, the blue (L-L) points (20 points) denoted zones where they have similarly low values clustered near each other so those points have low priority. Besides, the pink (H-L) points (20 points) and the green (L-H) points (17 points) were outliers that were points in which case we found a high value or high TA surrounded by low TA and conversely for the green points.

In the next step, the TA density map and the significant map of clusters were overlapped together as Figure 7. The statistical test process of clusters helps to hinder the

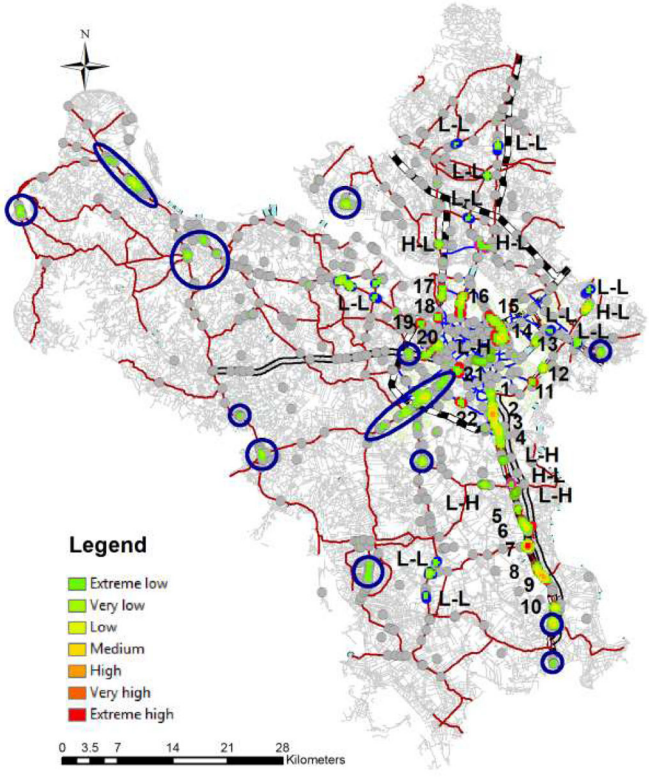


Figure 7. A hotspot priority map.

happening of too many clusters (a false positive). Figure 7 showed that there were several of the TA hotspots determined by the KDE method were not really dangerous. These locations were determined as the grey points. These locations were circled in blue as shown in Figure 7.

The locations are (L-L), (H-L), or (L-H) were noted in the map (Figure 7). In the next step, only the (H-H) clusters were considered for further analysis. The (H-H) clusters frequently concentrate on central districts and on National Highway 1 A (NH-1A) and were numbered from 1 to 22 on the map (Figure 7).

In this case, in order to identify the best arrangement of TA density in different classes, we apply the natural-breaks (Jenks) classification technique. The dangerous levels of these locations also were ranked and displayed visually on the map (Figure 7). These (H-H) clusters were ranked in descending order based the intensity of clusters (see the columns $(1-4)$ in Table 3).

However, to validate the proposed methodology, the obtained results were compared to the Gi* statistics and the EPDO method.

4. The validation of the results

4.1. The validation of the (H-H) clusters

The Gi* statistics was applied to validate the (H-H) clusters. The outputs include Gizscores and Gipvalues per features and presented in Table 3. The outputs showed that there were 18 sites with Gizscores > 2.58 and Gipvalues < 0.01. It indicates that these sites were (H-H) clusters with the statistical meaning at the confidence level of 99%. Besides, there were 4 sites (5, 18, 19, and 21) with Gizscores varied $(1.96-2.58)$ and Gipvalues < 0.05. It indicates that these sites were (H-H) clusters with the statistical meaning at the confidence level of 95% (see the columns $(5-8)$ in Table 3). Thus, the results from the proposed method were satisfied the validation process in terms of the (H-H) clusters.

4.2. The validation of ranking

The EPDO method was applied to validate the ranking of hotspots. The EPDO scores were summarized in column (9) in Table 3. Table 3 shows that the orders of hotspots were determined by both these methods are quite similar. However, there were three different cases including sites (5, 12, and 21). Thus, there were 19/22 sites satisfying the validation process in terms of ranking, accounting for 86%. Thus, the gained results from the proposed method are reliable and exact.

5. Traffic accident preventive strategies

Table 4 shows the top 10 most dangerous locations, of which four locations are uncontrolled crossings (locations No. 7, 9, 2, and 10). These locations were formed from the illegally opened roads as mentioned above. These locations were extremely dangerous and ranked as the highest priority to be handled first. The results of the proposed methodology also were consistent with the TA statistical reports in the (2015-2017) period in Hanoi and the observations from reality. From there, we suggested several countermeasures corresponding to each location.

6. Conclusions

The study proposed a procedure that determined TA hotspots by using a GIS-based KDE algorithm, then evaluated the statistical significance of the hotspot clusters by using spatial autocorrelation analysis (global and local Moran's I statistics), and finally organized according to their importance levels. The approach was tested by using TA dataset during three years (2015-2017) in Hanoi.

Especially, the proposed approach took into account SI in determining the hotspots, which was often ignored in past studies. Moreover, the study was carried out on a large scale (macro-analysis) that included the entire road network such as local and major roads, downtown collector and residential streets, urban highways, railways, and bridges, etc. while past studies often focused on one type of road such as highways or major roads. The results showed that this approach enables traffic authorities to quickly and effectively determine the most hazardous locations with statistical significance at large scale areas. This is also a contribution of this study.

Currently, there is a debate about the use planar or network KDE in urban areas. Therefore, the results of this approach can confirm that the planar KDE is suitable for applying in urban areas which have dense road networks with dispersed TA. This is also a contribution of this study.

Table 3. The (H-H) clusters ranked in descending order.

				(H-H) Cluster Validation				Ranking Validation	
		Intensity		GiZScore					
Rank (1)	Site ID* (2)	Strength (3)	Level (4)	Range (5)	> 2.58 SD (6)	GiPValue (7)	CL (%) (8)	EPDO scores (9)	R (10)
1	7	1006.12	Extreme high	3.00-3.50	☑	0.00089	99	730	☑
2	9	682.60	High	3.00-3.50	☑	0.00023	99	650	☑
3	2	599.39	High	3.00-3.50	☑	0.00059	99	640	☑
4	8	521.62	Medium	3.00-3.50	☑	0.00034	99	520	☑
5	3	433.90	Medium	3.00-3.50	☑	0.00675	99	520	☑
6	14	398.33	Medium	3.00-3.50	☑	0.00079	99	440	☑
7	12	396.37	Medium	3.00-3.50	☑	0.00025	99	360	X
8	6	392.38	Medium	3.00-3.50	☑	0.00358	99	400	☑
9	10	365.00	Medium	3.00-3.50	☑	0.00677	99	360	☑
10	16	335.60	Low	2.58-3.00	☑	0.00693	99	320	☑
11	20	321.68	Low	2.58-3.00	☑	0.00465	99	320	☑
12	13	319.33	Low	2.58-3.00	☑	0.00123	99	320	☑
13	1	282.19	Low	2.58-3.00	☑	0.00012	99	280	☑
14	4	271.03	Very low	2.58-3.00	☑	0.00197	99	280	☑
15	15	261.41	Very low	2.58-3.00	☑	0.00044	99	240	☑
16	5	235.90	Very low	1.96-2.58	X	0.04304	95	210	X
17	17	225.45	Very low	2.58-3.00	☑	0.00650	99	240	☑
18	22	210.90	Very low	2.58-3.00	☑	0.00788	99	240	☑
19	11	202.09	Very low	2.58-3.00	☑	0.00304	99	200	☑
20	21	189.01	Very low	1.96-2.58	X	0.03689	95	240	X
21	18	177.60	Very low	1.96-2.58	X	0.03227	95	121	☑
22	19	167.68	Extreme low	1.96-2.58	X	0.04501	95	120	☑

*ID corresponds to the locations on the map in Figure 7.
Note: ☑: Inside interval; X: Outside interval; CL: Confidence Level.
SD: Standard Deviation; R: Result.

Importantly, the study proposed the validation process for the results by applying the Gi* statistics method to validate the (H-H) clusters and the EPDO method to validate the ranking of hotspots. The Gi* method was selected because it can determine whether features with high values or low values tend towards cluster. The results from the proposed method were satisfied the validation process in terms of the (H-H) clusters. The EPDO method was chosen because it accounts accident SI in identifying hotspots and does not require traffic volume data. The results showed that the orders of hotspots were determined by the proposed approach and the EPDO method was quite similar. However, there were three different cases including sites (5, 12, and 21) (see Table 3). Thus, there were 19/22 sites satisfying the validation process in terms of ranking, accounting for 86%. Thus, the gained results from the proposed method are reliable and precise. To the best of my knowledge, this is the first study in which these methods were applied together to bring about the accuracy outputs. The procedure detailed the steps so that traffic authorities can apply and implement easily and effectively. This is also a significant contribution of this study.

Another significant benefit of this study is that typically we need to compare all sites across the entire study area to determine hotspots when applying traditional methods (network screening methods). This takes time and effort. Instead, after identifying the potential hotspots with statistical significance by applying the proposed approach, we only need to apply traditional methods on these potential hotspots.

The statistical test process of clusters helps to hinder the happening of too many clusters that were determined by the KDE method because they were not really dangerous (a false positive). Besides, the study determined also the most dangerous locations (H-H) clusters with statistical significance, organized them according to their significant levels, and suggested corresponding preventive strategies. In addition, the results also show the locations of four types of

Table 4. The top 10 most hazardous locations and TA preventive strategies.

Rank	Site ID[a]	Prior level	Location	Accident reason	Suggestion
1	7	Extreme high	Cho Tia station zone	Lack of safe equipment owing to illegally opened roads; narrow road; lack of median; wrong lane; crowded area, crossroads.	Remove the illegal crossroads, or add roadside safe equipment like lights, signals, median (removable poles), crosswalks, speed limit signs; or widening the road, etc.
2	9	High	Van Diem station zone	Lack of safe equipment owing to illegally opened roads; narrow road; lack of median and crosswalk; wrong lane; crowded area; illegal overtaking; over-speed; crossroads.	Remove the illegal crossroads, or add roadside safe equipment like lights, signals, median (removable poles), crosswalks, speed limit signs; or widening the road, etc.
3	2	High	Van Dien station zone	Lack of safe equipment owing to illegally opened roads; narrow road; lack of crosswalk; crowded area; over-speed; crossroads.	Remove the illegal crossroads, or add roadside safe equipment like lights, signals, crosswalks, speed limit signs; or widening the road, etc.
4	8	Medium	Do Xa, Thuong tin	Illegal overtaking, over-speed, crossroads.	Law enforcement, education, speed limit signs.
5	3	Medium	Ngoc Hoi road (Km 12 - NH-1A)	Passing red light, high speed, crossroads, out of control, wrong lane.	Law enforcement, education, speed limit signs.
6	14	Medium	Trang Tien	Passing red light, over-speed, out of control, illegal turning.	Law enforcement, education, speed limit signs.
7	12	Medium	Thanh Tri Bridge	Over-speed, out of control, bad road surface.	Law enforcement, education, engineering improvement.
8	6	Medium	Quat Dong, Thuong Tin	Wrong lane, illegal overtaking.	Law enforcement, education, speed limit signs.
9	10	Medium	Phu Xuyen station zone	Lack of safe equipment owing to illegally opened roads; narrow road; lack of median; wrong lane; crowded area, crossroads.	Remove the illegal crossroads, or add roadside safe equipment like lights, signals, crosswalks, speed limit signs; or widening the road, etc.
10	16	Low	Xuan La – Vo Chi Cong	Wrong lane, unsafe distance, over-speed.	Law enforcement, education.

[a]ID corresponds to the locations on the map in Figure 7.

correlation, including (H-H), (L-L), (H-L), and (L-H) clusters with statistical significance and the levels of dangerous also were illustrated in a map visually. These hotspots were statistically significant meaning that they were very far from the random arrangement. Especially, there were several dangerous locations, formed from the illegally opened roads, were also determined (locations No. 7, 9, 2, and 10 in Figure 7 and Table 4). These locations were extremely dangerous and ranked as the highest priority to be handled first. The results of the proposed methodology also were consistent with the TA statistical reports in the (2015-2017) period in Hanoi and the observations from reality.

The results showed that the approach was effective and precise in identifying TA hotspots in Hanoi. The integration of the KDE method and spatial autocorrelation analysis helps to overcome the drawbacks of the KDE method. This method can identify dangerous segments in which many accidents happened in cases of even distributions. From there, the determination of TA hotspots will be improved with high accuracy. These outcomes will not only enable traffic authorities to comprehensively understand the reasons for each accident but also to help them manage and deal with hazardous areas according to the prior order in case of limited expenses and allocate traffic safety sources suitably. Thus, it is necessary to evaluate the statistical significance of hotspots.

The limitation of this study is that several variables such as traffic exposures, environment, land use, geometric conditions of roads, and weather conditions were not considered in this analysis. Therefore, it is also a recommendation for further studies.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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